Aditya Singh's proposal:
Value leakage is a paper from Owain Evans' group where models bias towards outcomes they prefer. Section 3 contains the *Donation Bet* experiment, where models are asked to perform a Fermi estimate for the number of giraffe spots in the world. However, the user says they will donate to a good/bad cause depending on if the estimate is above/below threshold, and models will motivated reason their way towards the good outcome, even when claiming in their thoughts that they are being unbiased.

- Try to better understand what motivated reasoning looks like. Should we think of this as unfaithful CoT?
- I think sentence resampling would be valuable here, but it takes a long time. If you do this, you can exclude it from the 5-hour limit
- I was able to replicate the behavior, you can start from this codebase: https://github.com/adsingh-64/value-leakage
- Below is my replication on different models, if you look at Qwen 3.5 122B A10B, it might be interesting to explore with the J-lens for it here

The problem I want to pick here:

- Can we take the result from the Value Leakage paper and try to understand why models do motivated reasoning?
- Can we attribute different rollouts with motivated reasoning to being "model consciously does this" vs "model subconsciously does this"?
    - J-lens here (how to operationalize this is a difficult question though)
- In rollouts without verbalized motivated reasoning, can NLAs be useful?
- Can we design a suite of categories in the Donation Bet setting and see how motivated reasoning changes?
- Can we find a linear direction like "do I wish to support charity" and see if it fires and causally mediates this?
- Can we finetune the model to have longer CoTs and see if the unfaithfulness goes away or the opposite, finetune it to write a smaller CoT and see if the model becomes more unfaithful?